

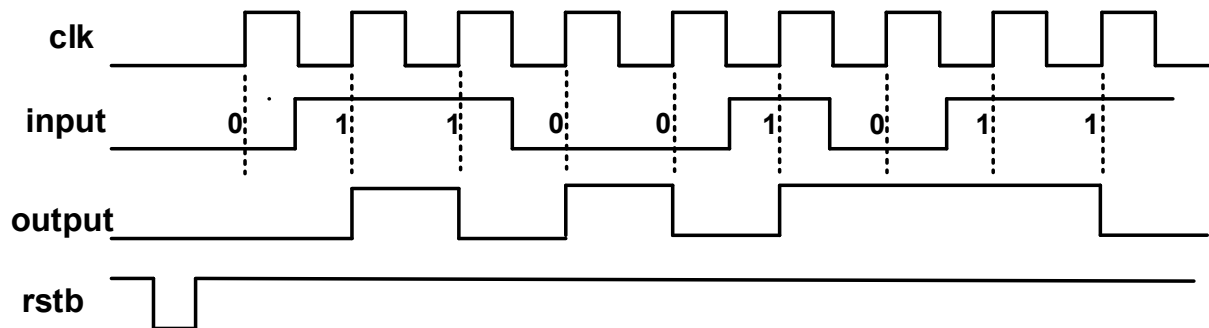
Assignment #3
Due by 11:59pm on May 24
ECE 303: Advanced Digital Logic Design

Problem 1. FSM Design and Verilog Coding

Design a Finite-State-Machine circuit realizing transition detection of input signal described below:

Input bit streams, e.g. 011001011 are received one bit per clock cycle as shown in waveforms below.

Whenever a 0 to 1 transition or 1 to 0 transition at input is detected, the output should remain 1 for one clock cycle and return to 0. If continuous transitions are detected, the output should remain 1 for the transition cycles until all transitions are finished.



- (1). Please derive the state diagram using Moore's FSM; No need for initial state (Assume after reset, the state machine start from the first state, e.g. S0 and the input remains at 0 until the first clock cycle after reset).
- (2). Please derive the transition table and assign binary codes to states;
- (3). From the transition table, derive the next state and output logic expression, and draw the resulting circuit schematic with simple logic gates.
- (4). Write RTL for this FSM and perform Xcelium simulation. Submit your RTL and testbench as well as snapshots of waveforms clearly indicating the correct functionality of the design. Highlight (cycle) the portion of the waveforms which show the proper functionality.
- (5) Perform synthesis with a clock period of 0.5ns. Report your SDC file, timing report and area report.

Problem 2. Verilog coding

- (1) Write a RTL code for a memory module. The inputs include clock signal “CLK”, write enable “WE”, reset signal “RSTB”, 8-bit input address “ADDRESS”, chip enable signal “CE”, 8-bit input data “INPUT”, 8-bit output data “OUTPUT”; A special sequence command “SEQ”;
 - (2) The operations are as follows:
 - At RSTB=0, all memory contents as well as internal registers are reset to 0.
 - “CE” signals need to be “1” to enable read or write operations; if “CE” is “0”, the “OUTPUT” should remain as 8’bz;
 - Total memory address is $2^8=256$;
 - If “WE” is 1 and chip is enabled, the “INPUT” value is being written into the memory at “ADDRESS”, OUTPUT value is equal to the “INPUT” value at the next clock cycle;
 - If “WE” is 0 and chip is enabled, the value of “ADDRESS” is being read from memory; OUTPUT value is equal to the memory content at “ADDRESS” at the next clock cycle;
 - When the special sequence command “SEQ”=1, the “ADDRESS” automatically add 1 at every additional clock cycle when “SEQ”=1; The read and write operates as normal operation. For example, when “SEQ”=1, the first cycle, the chip operate normally at given “ADDRESS”; If “SEQ” continuously to be 1, the read/write will happen at “ADDRESS+1” at the next cycle and then “ADDRESS+2” at the following cycle until SEQ=0 again.
 - (3) In addition to the RTL code, please also write a small test bench code to instantiate the module and test the module. Show at least two read and write operations when SEQ=0 and two read and write operations when SEQ=1. Assume a clock period of 1ns.
- Hint: memory can be created from defining a register such as “reg [7:0] MEM [0:255];”.
- (4) Write RTL for this memory module and perform Xcelium simulation. Submit your RTL and testbench as well as snapshots of waveforms clearly indicating the correct functionality of the design. Highlight (cycle) the portion of the waveforms which show the proper functionality.
 - (5) Perform synthesis with a clock period of 1ns. Report your SDC file, timing report and area report.